REVIEW 2 | 23CSE452 BUSINESS ANALYTICS | ADVANCED ANALYTICS: TEXT MINING AND TIME SERIES

Mobile App Update Impact and Failure Detection

What users complain about, what is coming next, and which app to fix first

1,137,987 Play Store reviews | 11 apps | 3 domains | 1 April to 20 September 2026

Team number: 10

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Live dashboard
team10-app-reviews.streamlit.app

Review 2 presentation: 20 marks

Three Business Questions, Two Unit 3 Methods

1 What are users complaining about?

Method 1, text mining: NMF topic models on 167,357 complaint reviews, VADER sentiment per topic, words that set each app apart.

2 How many complaints are coming, and was a release to blame?

Method 2, time series: ARIMA/SARIMA forecasts tested on a rolling backtest, unusual-day alerts, and before/after tests for 224 app versions.

3 Which app needs attention first, and what should it do?

Combined insights: six warning signs from every stage give each app a priority and 65 recommendations, all in a live dashboard.

How the work flowed: one stage after another, each merged through a pull request

R1

Review 1 data

1.14M reviews, 9 issue tags, VADER, classifier

1

Stage 1: preparation

text corpora, daily series, dashboard base

2

Stage 2: text mining

17 topics, sentiment, evaluation

3

Stage 3: forecasting

11 ARIMA models, backtest

4

Stage 4: evaluation

accuracy, intervals, alerts

5

Stage 5: impact + insights

205 releases tested, priorities

Text Mining: Preparing the Review Text for Topic Models

Two corpora

167,357

complaint reviews (1–2★) kept, 75.5% of 221,581

166,929

praise reviews (4–5★) kept as a contrast, 19.2% of 871,380

Kept: at least 3 content words after cleaning. "worst" or "good" alone cannot carry a topic.

Median length: 11 words for complaints, 4 for praise. 3★ reviews are left out.

Cleaning steps

1. Lowercase; remove links, e-mails, emojis, digits
2. Remove brand phrases ("google pay", "phone pe") so "pay" stays usable
3. Expand contractions: "didnt" → "did not"
4. Drop stop words but keep negations: "not delivered" keeps its meaning
5. Hinglish: "nahi", "nhi" → "not"; drop filler words ("hai", "mera")
6. Drop app names, so topics describe problems, not apps
7. Lemmatise (WordNet): "refunded" → "refund", "orders" → "order"

Example

Raw: "PLEASE STOP CHEATING US BY CHARGING DELIVERY CHARGES FOR ORDERS ABOVE 100..."

Clean: stop cheat charge delivery charge order people number fraud

Document-term matrices

Matrix	Reviews	Terms
Complaints · Food	82,603	17,448
Complaints · Shopping	71,413	18,665
Complaints · Payments	13,341	2,982
Praise · Food	74,647	6,875
Praise · Shopping	75,672	7,097
Praise · Payments	16,610	1,763

Words and two-word phrases ("not deliver", "customer care") in at least 10 reviews and at most half of them.

Time Series: Daily Complaint Counts, Data Gaps and Releases

1 The series

Daily count of 1–2★ reviews per app: 11 series of 173 days. Counts, not shares: the late-April feed gap removed mostly positive reviews, which distorts shares far more than counts.

2 Data-quality flags

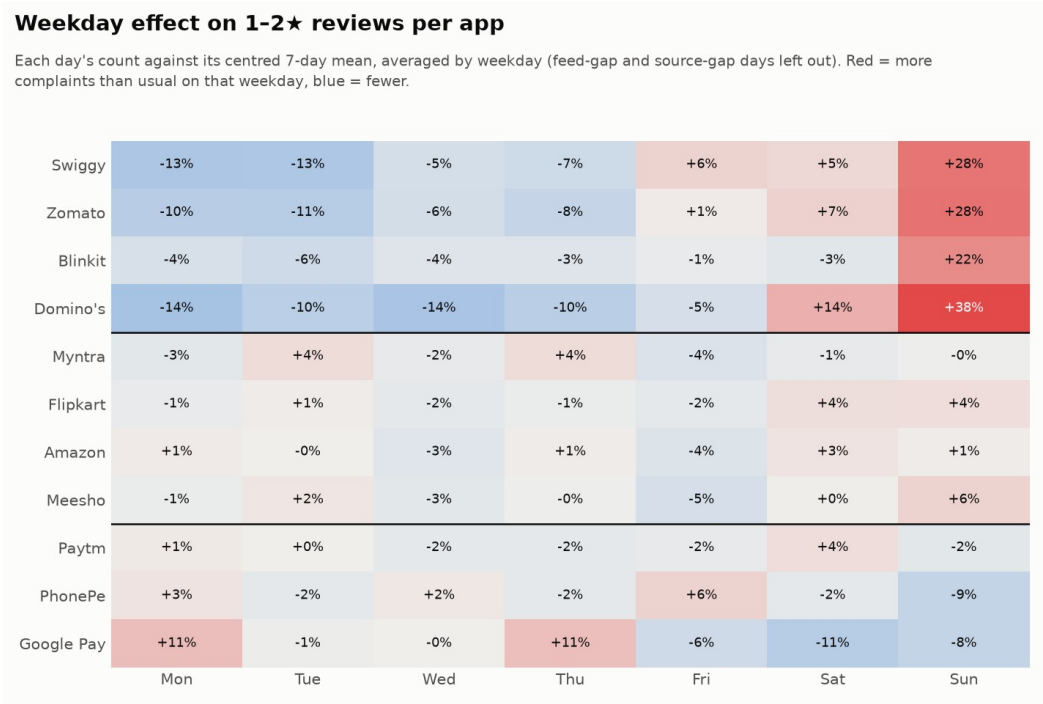
Feed gap 21 Apr – 5 May for five apps becomes a 0/1 input to ARIMA. Zomato's missing day (24 Jul) is filled with the same weekday one and two weeks either side.

3 224 release events

From 2,507 app versions: at least 30 reviews, first seen after 1 April, and numbered higher than every earlier version (old builds removed).

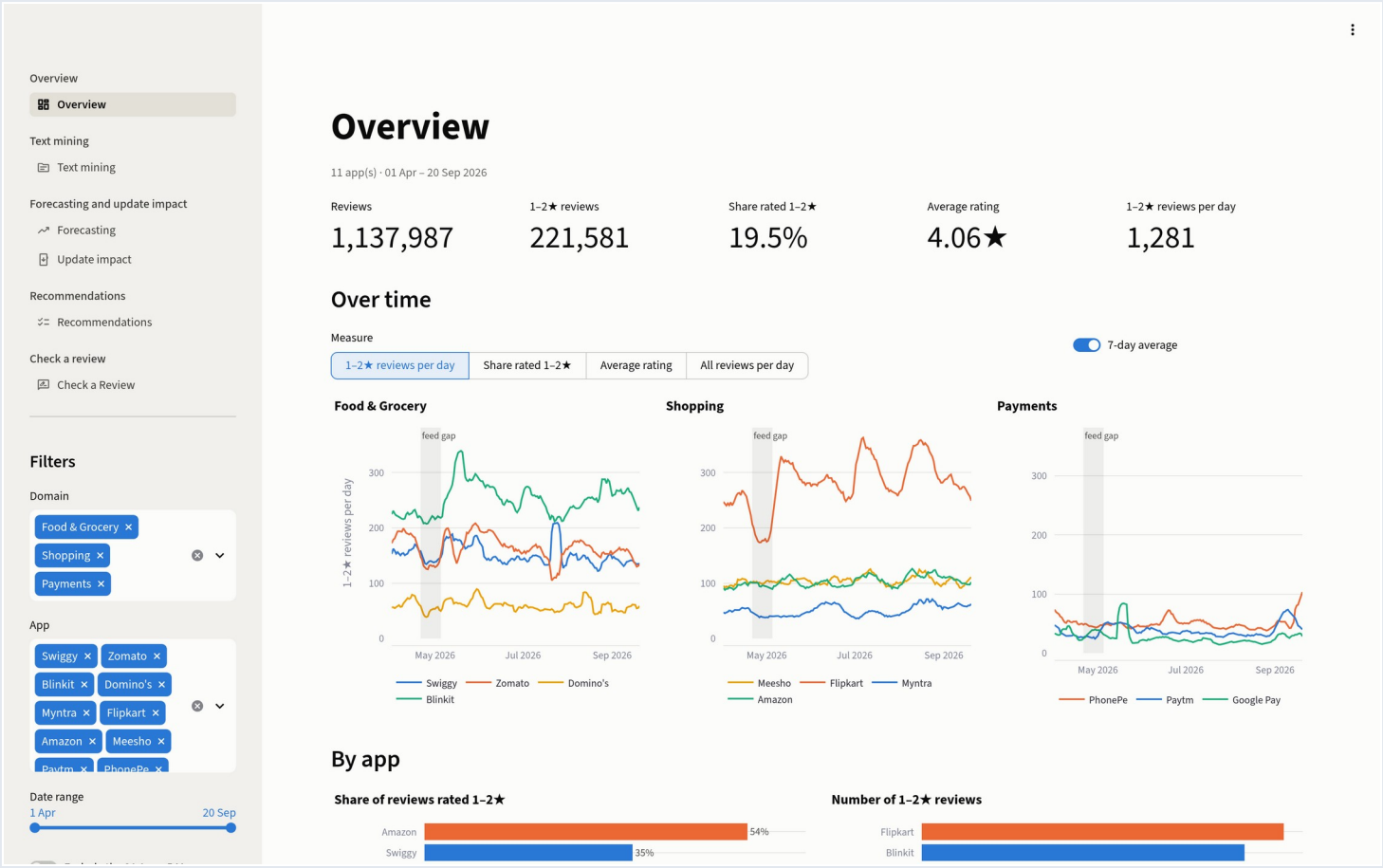
4 Checks for ARIMA

ADF test: 9 of 11 apps stationary in level, so d = 1 only for Zomato and Blinkit. STL seasonal strength below 0.64 everywhere, so D = 0.



Weekly pattern: food apps get 22–38% more complaints on Sundays; shopping apps have no weekday pattern. This is why only food apps get seasonal ARIMA terms.

Dashboard: One App, Six Pages, Live on Streamlit Cloud



Built in Stage 1

- Shared filters on every page: domain, app, date range, exclude the feed gap
- Reads small precomputed tables, so it opens in seconds
- Overview: volume, 1–2★ share, trends per domain, top issues

Six pages

Page	Built by
Overview	Aditya
Text mining	Akshay
Check a review	Harshini
Forecasting	Kanishka
Update impact	Kanishka
Recommendations	Kanishka

team10-app-reviews.streamlit.app

Redeploys from GitHub after every merge.

Topic Modelling: 17 Complaint Topics with NMF

How the topics were found

TF-IDF + NMF: the counts become TF-IDF weights; NMF splits them into topics (V ≈ W × H), each a weighted list of terms.

Why NMF: on very short reviews LDA mixes themes; NMF with nndsvda start is deterministic and runs in seconds on 167,357 reviews.

One model per domain: K = 6 Food & Grocery, 6 Shopping, 5 Payments (chosen by coherence, next slide).

Per review: its dominant topic plus VADER sentiment, so every topic has a mean score and a share of strongly negative reviews (≤ −0.5).

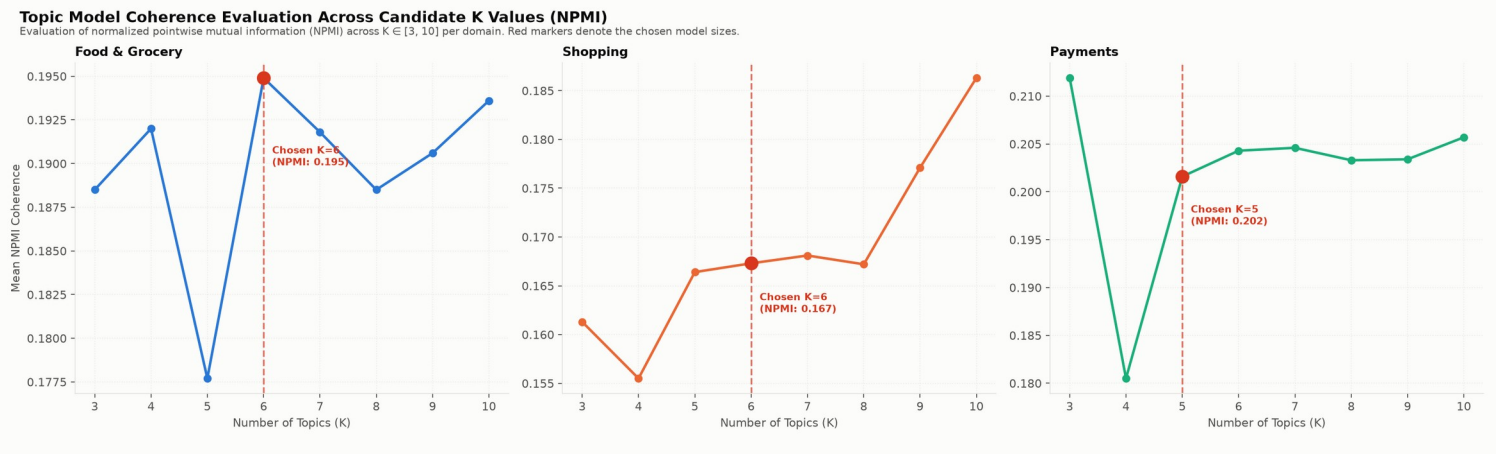
Contrast: a second NMF (K = 4 per domain) on 4–5★ reviews shows what satisfied users praise.

Reused: each pipeline is saved, so the dashboard can give a new review its topic.

Domain	Complaint topic	Share	Mean VADER	Strongly neg.
Food & Grocery	Order Cancellation & Refund	35%	-0.27	36%
	Customer Support & Bot Loop	19%	**-0.40**	56%
	Excessive Delivery & Handling Charges	11%	-0.21	24%
	Poor Food Quality & Stale Items	13%	**-0.44**	59%
	Delivery Delays & Rider Tracking	18%	-0.29	39%
	COD & Payment Option Failures	5%	-0.12	18%
Shopping	Return Rejection & Defective Products	25%	-0.23	33%
	Customer Service & Escalation Failure	18%	-0.24	41%
	Courier Delays & Delivery Agent Issues	16%	-0.29	41%
	Poor Shopping Experience & Fabric Quality	9%	**-0.59**	81%
	Extreme Dissatisfaction & Scam Allegations	10%	**-0.61**	80%
	Order Cancellation & Rescheduling	22%	-0.31	41%
Payments	Failed Transfers & Debited Amounts	52%	-0.15	27%
	App Crashes & Update Regressions	9%	-0.09	16%
	Device Environment & Security Scan Errors	10%	-0.11	18%
	Account Blocking & AutoPay Issues	10%	-0.13	21%
	Customer Support Unresponsiveness	18%	-0.23	40%

Share = share of the domain's topic-assigned complaints. Red = mean sentiment of −0.4 or lower.

Evaluation: Coherence, Keyword Tags and the Untagged 43%



Choosing K (NPMI coherence)

K = 3 to 10 tried for each domain.

Food & Grocery peaks at K = 6 (0.195).

Shopping and Payments are almost flat, so we took the K where every topic is distinct and can be named; larger K split delivery and support into near-duplicates.

1 Topics agree with the keyword tags

The customer-support topics carry the Customer Support tag in 71% (food) and 76% (shopping) of their reviews.

Delivery-delay topic: 51% Delivery Delay tag. Device-environment errors: 46% Crash & Stability.

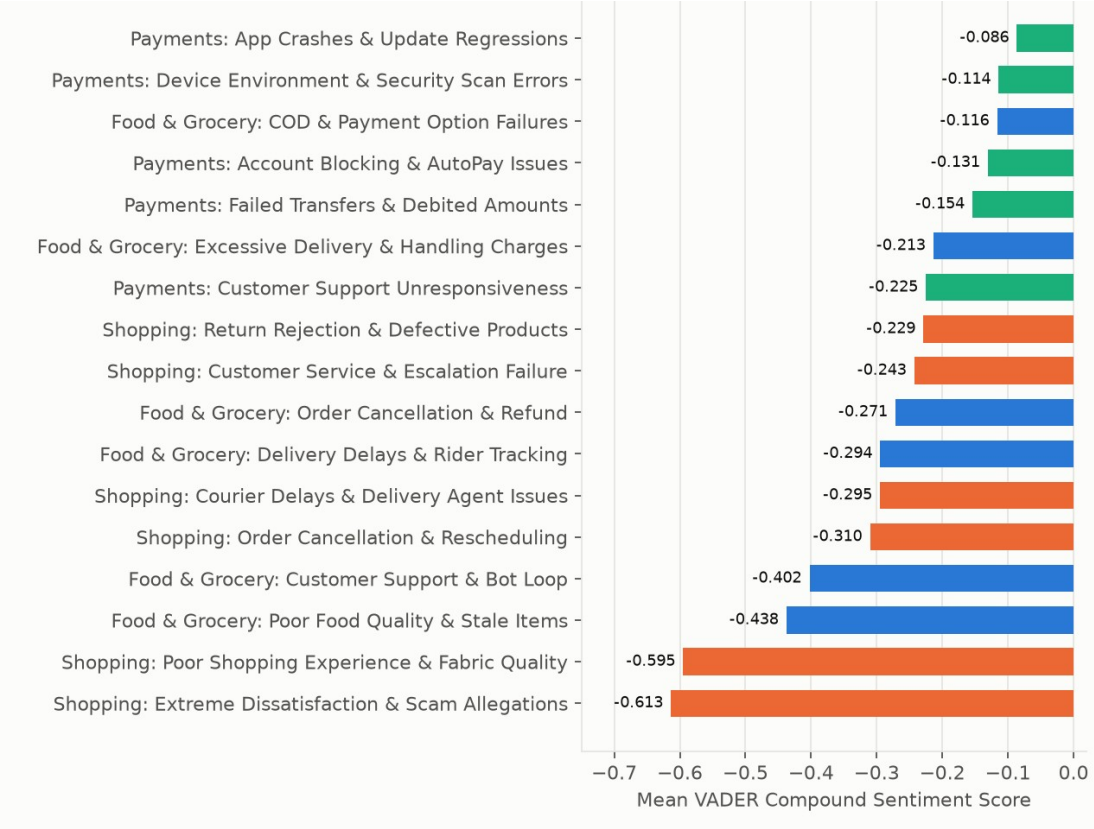
2 They also find what keyword rules miss

71,929 of 167,357 complaints (43%) match none of the 9 keyword tags. The topics place each one: order cancellation and refund 13,073, returns 8,275, food quality 6,089.

Untagged share is highest for COD failures (80%) and failed transfers (72%): Hinglish and informal wording.

Interpretation: The Angriest Complaints and How Apps Differ

Mean VADER sentiment of each complaint topic



Loss and broken trust are angriest

Shopping scam allegations −0.61 and poor quality −0.60 (about 80% strongly negative). Stale food (−0.44) angers more than late food (−0.29).

Technical faults are reported calmly

Payment app crashes −0.09, device-environment errors −0.11: users describe them factually, so tone is not a measure of severity.

Each app has its own complaint words

Swiggy: cancellation fee. Zomato: Gold, platform fee. Blinkit: damaged items. Myntra: exchange, size. Flipkart: Minutes, reschedule. Amazon: Prime membership. Meesho: account verification. Paytm: "device environment". PhonePe: charges. Google Pay: rewards.

What satisfied users praise

Fast delivery (food), product quality and price (shopping), easy UPI payments.

ARIMA Forecasting with a Rolling-Origin Backtest

Three models, refitted at every step

ARIMA / SARIMA (main): weekly seasonal terms for the four food apps; the feed-gap flag as an extra input for five apps (SARIMAX); 80% and 95% intervals.

Seasonal naive (baseline): same weekday last week.

Holt-Winters: damped trend + weekly season.

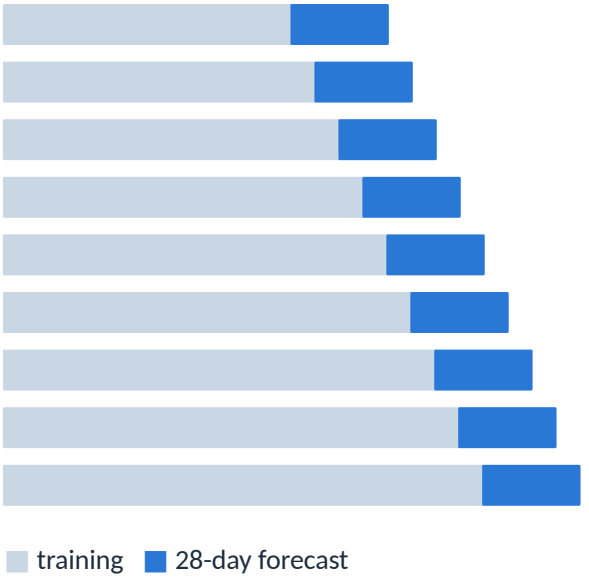
Choosing the orders

- d from the ADF test; D = 0
- ACF/PACF propose p, q ≤ 3 and lag-7 seasonal terms
- Lowest AICc wins (378 fits); within 2 points, the simpler model
- No look-ahead: backtest orders use the first 84 days only

App	Final order	Gap input
Swiggy	(0,0,1)(1,0,1)	yes
Zomato	(2,1,1)(1,0,1)	
Blinkit	(1,1,1)(1,0,1)	yes
Domino's	(0,0,1)(1,0,1)	yes
Myntra	(3,0,2)	
Flipkart	(1,0,1)	yes
Amazon	(1,0,1)	yes
Meesho	(0,0,2)	
Paytm	(2,0,2)	
PhonePe	(1,0,0)	
Google Pay	(1,0,0)	

(p,d,q)(P,D,Q) with a 7-day season. Only the food apps need seasonal terms, matching their Sunday peak.

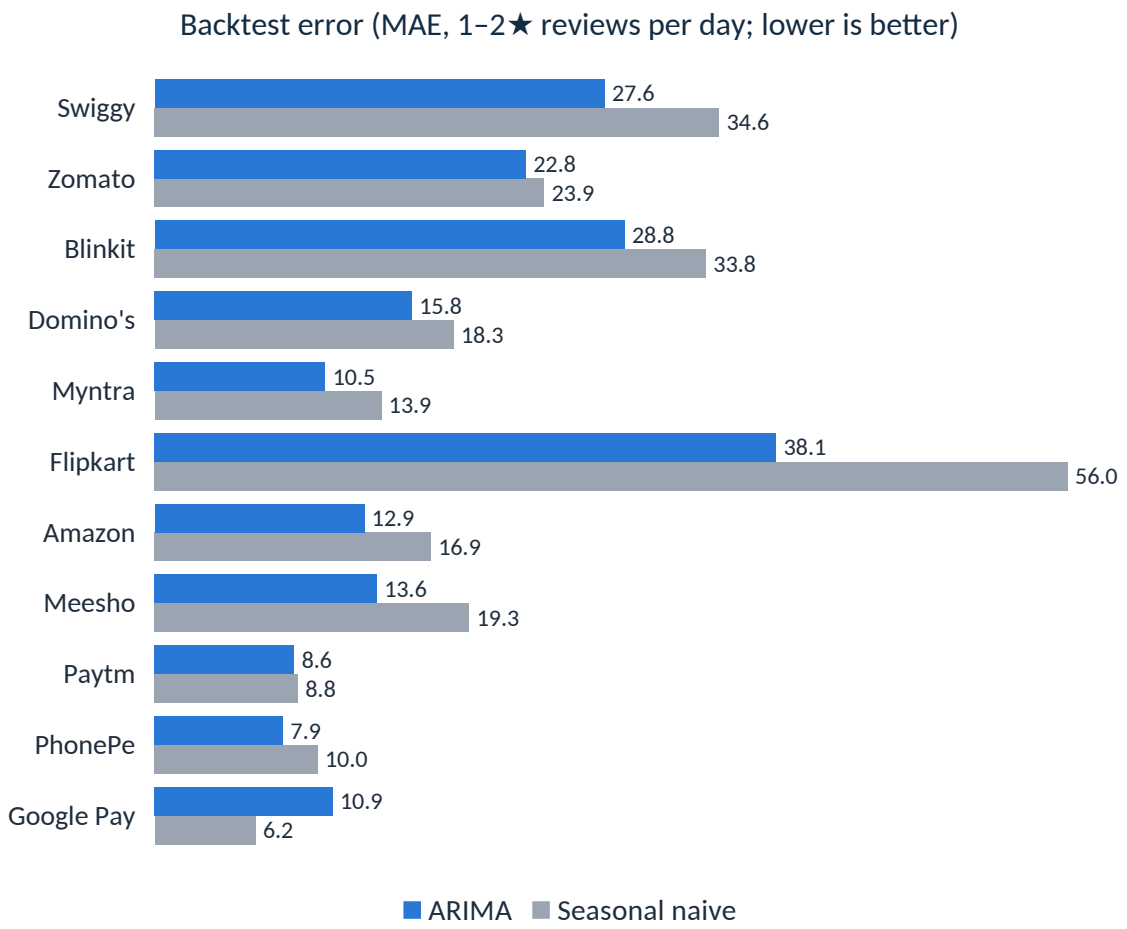
Rolling-origin backtest



First 84 days to train, then forecast 28 days; move one week and refit.

9 origins (23 Jun – 18 Aug) = 252 forecasts per app and model.

ARIMA Beats the Baseline for 10 of 11 Apps



10 / 11
apps where ARIMA has a lower error than the baseline

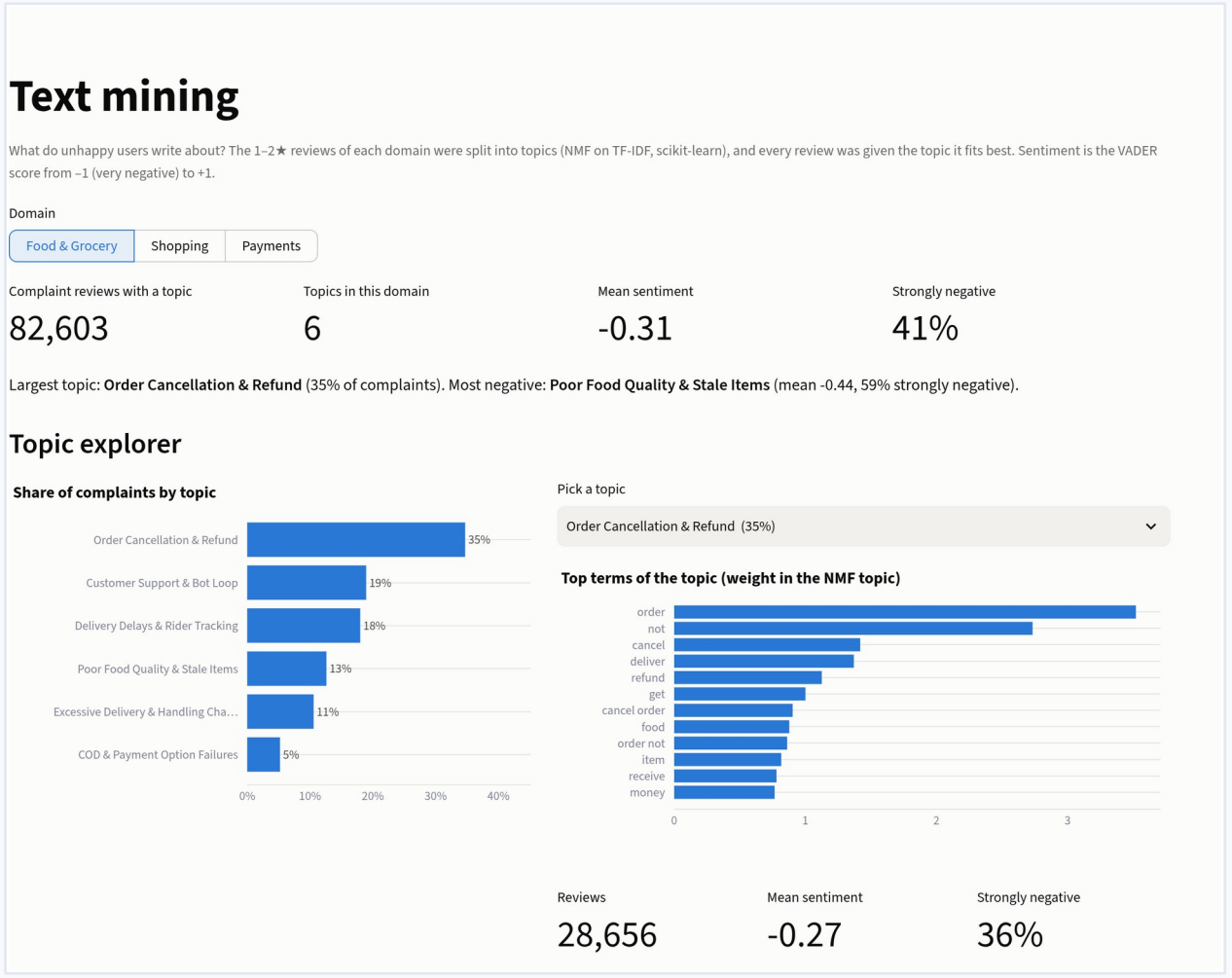
−32%
Flipkart error vs the baseline, the largest gain

1 / 11
Google Pay: the baseline wins (6.2 vs 10.9)

Forecast for 21–27 Sep (1–2★ reviews, week 1)

App	Forecast	80% interval	vs last 4 wks
Flipkart	1,830	1,596 – 2,056	−7%
Blinkit	1,711	1,488 – 1,929	−8%
Swiggy	1,037	889 – 1,180	+7%
Zomato	977	811 – 1,151	−7%
Meesho	737	676 – 801	+4%
Amazon	714	642 – 786	−6%
Domino's	413	327 – 495	+7%
Myntra	371	318 – 424	−10%
PhonePe	348	270 – 431	−10%
Google Pay	192	105 – 310	+3%
Paytm	136	75 – 205	**−61% (unreliable)**

Dashboard: Text-Mining Page



What a manager can do here

1. **Topic explorer:** pick a domain and a topic: its share, top terms, sentiment and real example reviews (189 readable examples, 3 per topic and app)
2. **Keyword-tag overlap:** which keyword tags the topic's reviews carry, and how many carry none
3. **Topics by app:** heatmap of each topic's share, or its share of strongly negative reviews
4. **Topic share over time:** monthly trend, following the date filter
5. **Distinctive terms:** the words that set one app apart from its domain

Forecast Evaluation: Accuracy, Intervals and Residuals

Accuracy over the backtest (mean of 11 apps)

Model	MAE	RMSE	MASE	Beats naive
ARIMA	**17.95**	**24.95**	**0.88**	**10 of 11 apps**
Holt-Winters	19.14	26.66	0.93	–
Seasonal naive	21.98	31.07	1.08	baseline

MASE below 1 = more accurate than a seasonal-naive forecast on the training data.

81% / 93%

of 2,772 forecasts fell inside the 80% / 95% intervals: close to nominal, 95% slightly narrow

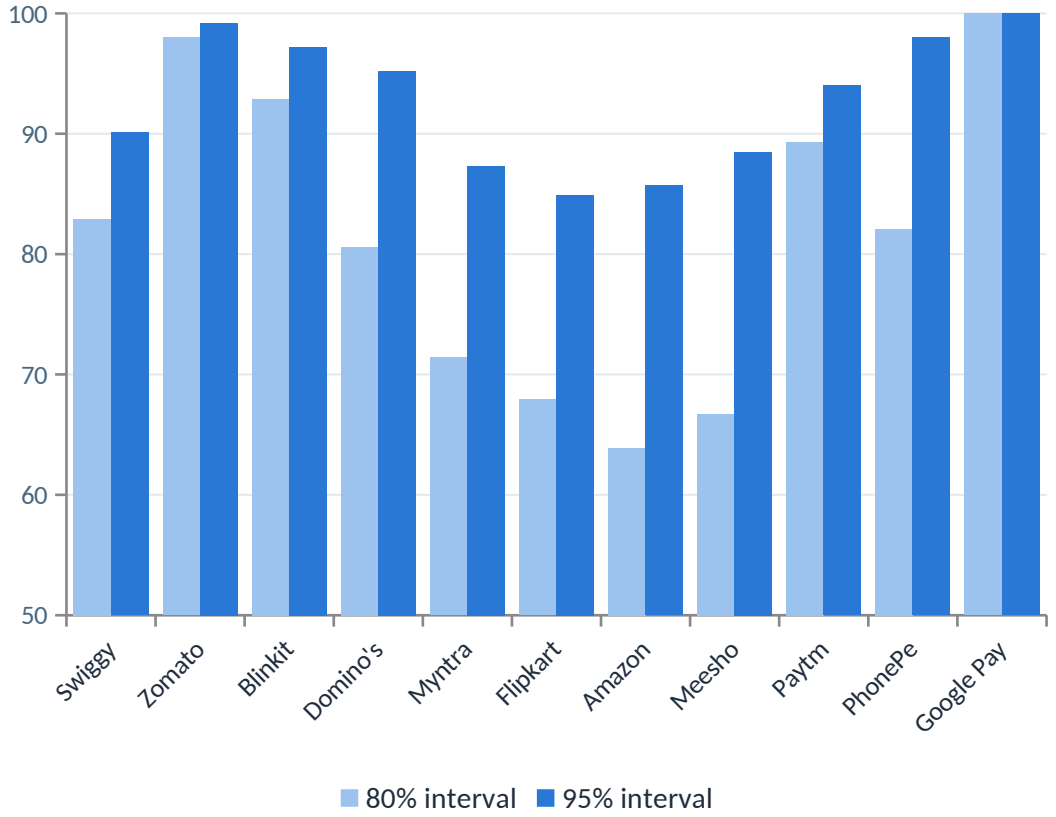
Rolling windows overlap, so this is a descriptive check, not an independent test.

11 / 11

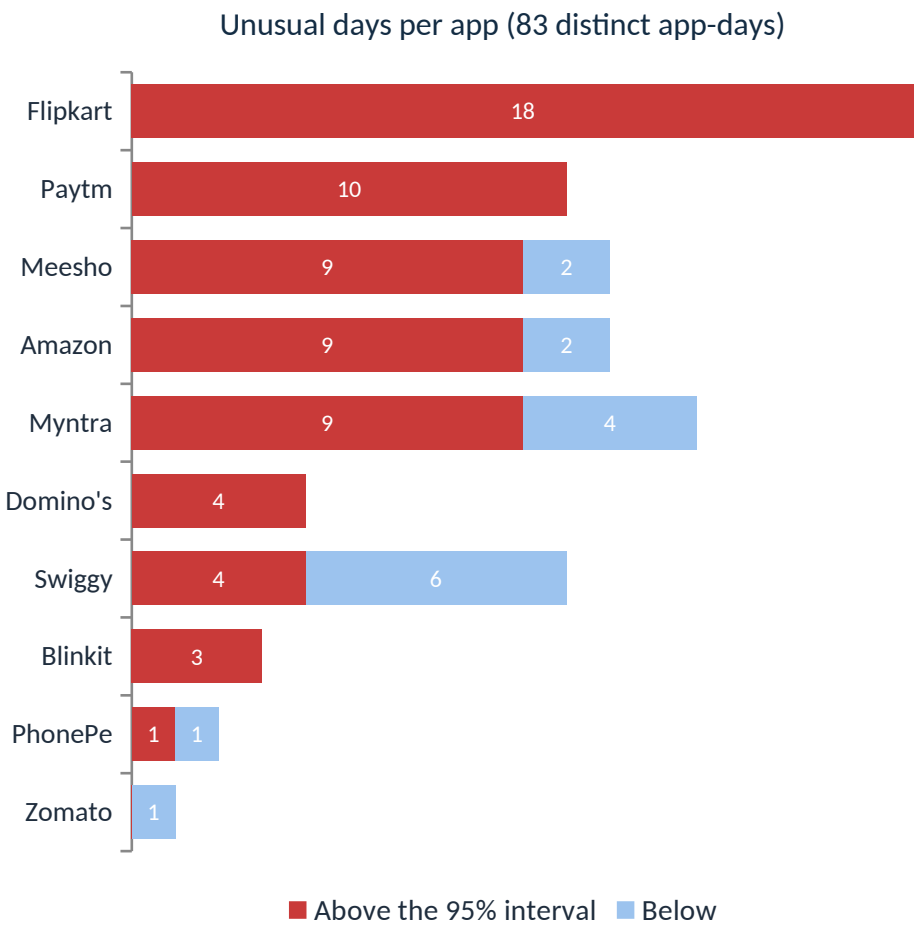
apps pass the Ljung–Box test at lag 14 ($p > 0.05$): no autocorrelation left in the residuals

Passing does not prove the residuals are random; it finds no pattern at that lag.

Share of backtest days inside the ARIMA interval (%); nominal 80 and 95



Unusual Days, and Which Apps Can Be Predicted



Alerts

A day is flagged when its count falls outside the 95% interval: 201 flagged forecasts = 83 distinct days (67 above, 16 below). Largest: Swiggy 24 Jul, 469 vs 157 expected; Flipkart 4 Jul, 451 vs 264.

Predictable: the food apps

A steady weekly rhythm. Zomato's and Blinkit's 95% intervals hold 99% and 97% of days.

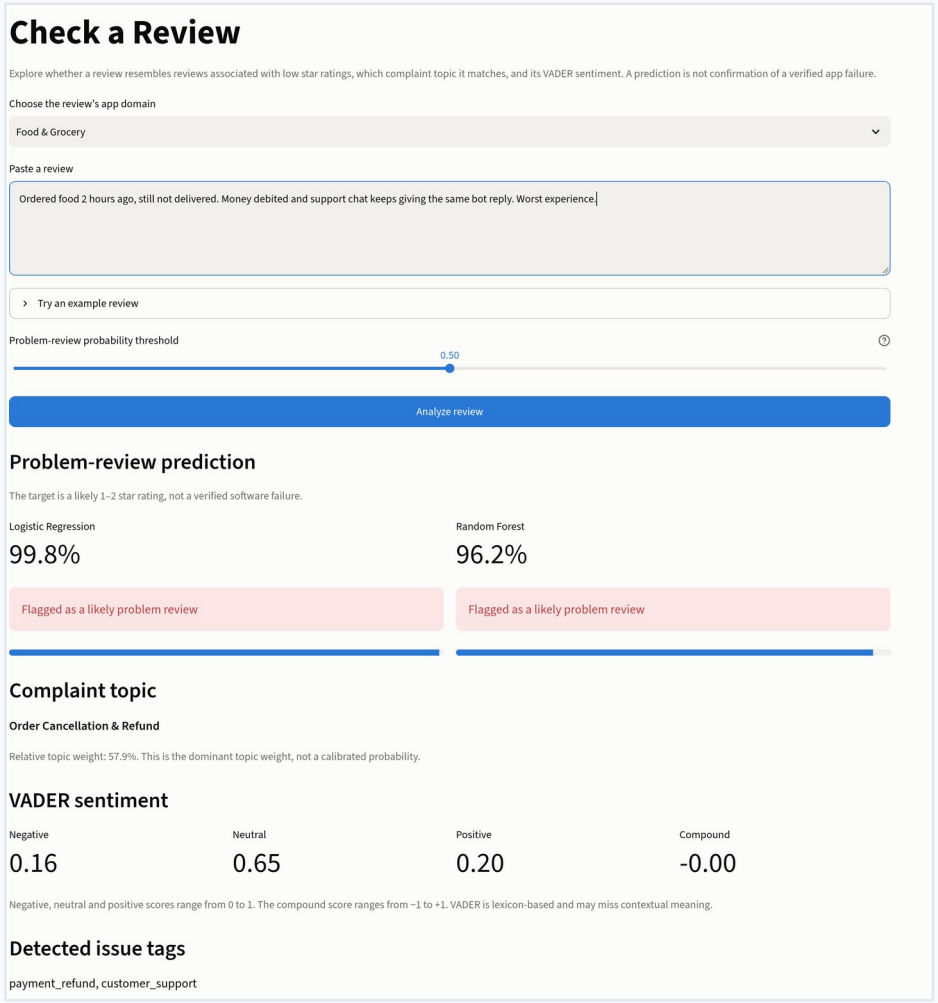
Hard: shopping apps with level shifts

Flipkart, Amazon, Meesho, Myntra: the 80% interval holds only 64–71% of days, and they have the most alerts.

Use with care: Google Pay and Paytm

Google Pay is low and noisy (the baseline wins). Paytm jumped to about 80 a day in September, so its week-1 forecast (−61%) is not reliable.

Dashboard: Check a Review



Paste any review and get four answers

1. **Problem review:** probability from the Review 1 Logistic Regression and Random Forest, with an adjustable threshold
2. **Complaint topic:** from the saved NMF pipeline of the chosen domain
3. **Sentiment:** VADER negative, neutral, positive and compound scores
4. **Issue tags:** the Review 1 keyword tags it matches

In the example shown

"Ordered food 2 hours ago, still not delivered. Money debited..." is flagged by both models (99.8% and 96.2%), gets Order Cancellation & Refund, and the Payment & Refund and Customer Support tags.

Its VADER compound is 0.00: sentiment alone would miss this complaint, which is why the page combines all four.

Update Impact: Did a New Version Make Things Worse?

Two comparisons per release (7-day windows)

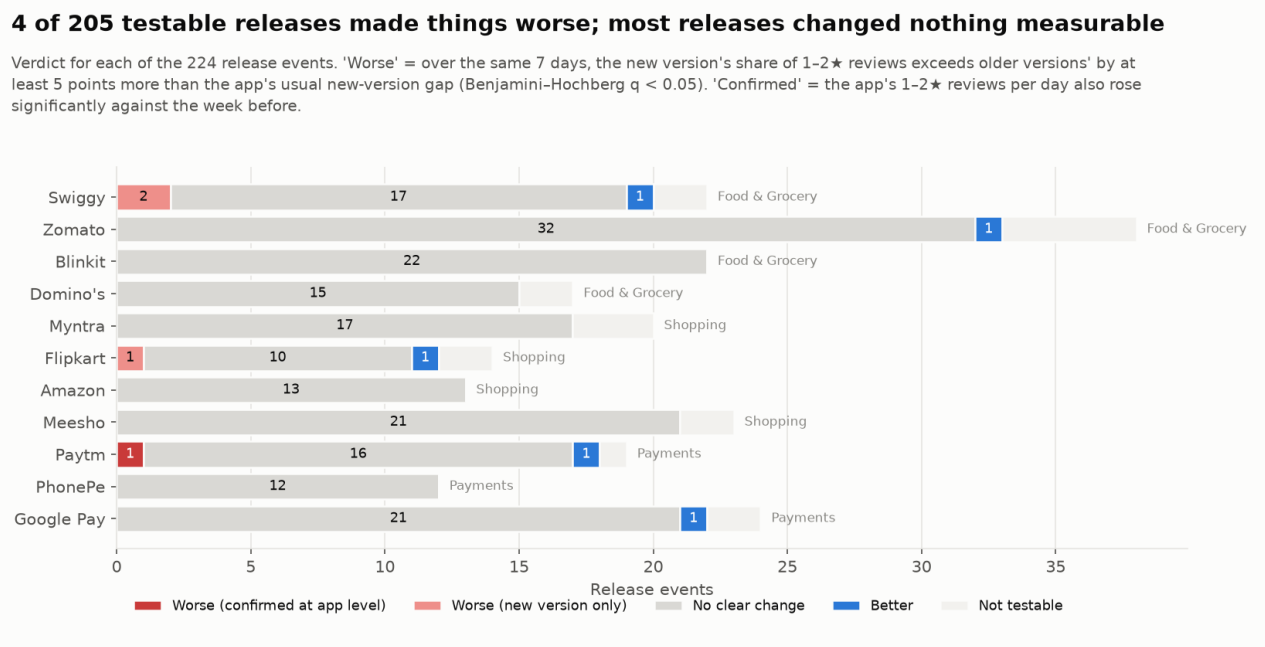
A. Before vs after: the app's 1–2★ reviews per day in the week before vs the week from the release (Welch t-test).

B. New vs older version (main): in the same 7 days, the 1–2★ share of users on the new version vs users still on older versions (two-proportion z-test). Same days, so sales and outages hit both groups.

Usual gap: new-version users on Amazon and Flipkart are always about 6 points more negative, so each release is compared with its app's usual gap.

Worse: at least 5 points beyond the usual gap, Benjamini–Hochberg q < 0.05 over all releases. Confirmed if A also rises.

Release day: the day the version got its 10th review; at least 30 reviews on each side, else not testable.



224
release events

205
testable

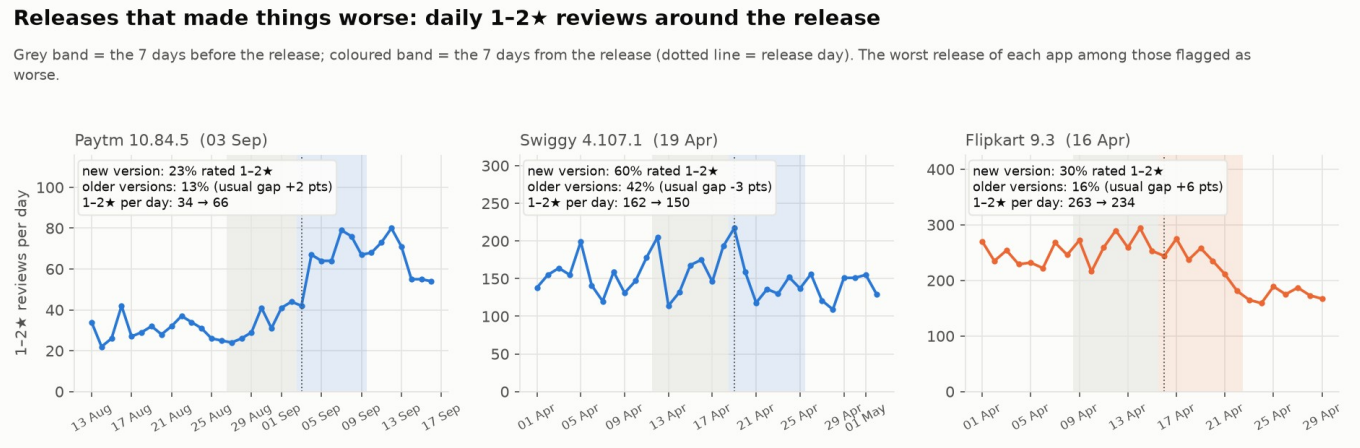
196
no clear change (96%)

5
better

4
worse, 1 confirmed

Four Releases Made Things Worse; Paytm 10.84.5 Is Confirmed

App	Version	Release	1–2★ new vs older	Beyond usual gap (pts)	1–2★ per day before → after	Issue that rose most	Verdict
Swiggy	4.107.1	19 Apr	60% vs 42%	+21.2	162 → 150	Customer Support +8.9	New version only
Swiggy	4.107.2	23 Apr	88% vs 83%	+8.5	163 → 137	–	New version only
Flipkart	9.3	16 Apr	30% vs 16%	+7.6	263 → 234	Customer Support +4.2	New version only
Paytm	10.84.5	3 Sep	23% vs 13%	+8.1	34 → 66	–	**Confirmed at app level**



How to read it

Paytm 10.84.5 is the clearest bad release: app-wide 1–2★ reviews nearly doubled (34 → 66 a day) and Harshini's forecast raised 6 alerts that week. It explains Paytm's September jump.

The other three have windows that overlap the late-April feed gap, so their shares are inflated.

No release made things worse for the other 8 apps. Same 4 flagged when the release day is the first review.

Combining Every Stage: Which App Needs Attention First

Six warning signs, one point each

1. **Many 1–2★ per day:** above the median app (103) time series
2. **High 1–2★ share:** above the all-app share (18.9%) time series
3. **Rising trend:** last 4 weeks +10% vs first 4 time series
4. **Forecast above recent:** reliable week-1 forecast +5% forecasting
5. **Days above forecast:** 5+ days above the 95% interval evaluation
6. **Release made it worse:** at least one worse release update impact

Score: 3+ High, 2 Medium, 0–1 Low. Every point traces back to one number from one stage.

Which apps need attention first: six warning signs from all stages

Filled dot = the sign applies. Score = number of signs (3+ High, 2 Medium, 0–1 Low). Signs: 1–2★ reviews per day above the median app; share of 1–2★ reviews above all apps; last 4 weeks 10%+ above the first 4; reliable week-1 forecast 5%+ above the last 4 weeks; 5+ days above the 95% forecast interval; a release that made things worse.

	Many 1–2★ per day	High 1–2★ share	Rising trend	Forecast above recent	Days above forecast	Release made things worse	Score	Largest complaint topic
Swiggy · Food & Grocery	●	●	○	●	○	●	4 High	Order Cancellation & Refund (37%)
Flipkart · Shopping	●	○	○	○	●	●	3 High	Return Rejection & Defective Products (25%)
Amazon · Shopping	○	●	●	○	●	○	3 High	Customer Service & Escalation Failure (24%)
Paytm · Payments	○	○	●	○	●	●	3 High	Failed Transfers & Debited Amounts (44%)
Blinkit · Food & Grocery	●	○	●	○	○	○	2 Medium	Order Cancellation & Refund (33%)
Meesho · Shopping	●	○	○	○	●	○	2 Medium	Return Rejection & Defective Products (28%)
Domino's · Food & Grocery	○	●	○	●	○	○	2 Medium	Order Cancellation & Refund (41%)
Myntra · Shopping	○	○	●	○	●	○	2 Medium	Return Rejection & Defective Products (26%)
Zomato · Food & Grocery	●	○	○	○	○	○	1 Low	Order Cancellation & Refund (33%)
Google Pay · Payments	○	●	○	○	○	○	1 Low	Failed Transfers & Debited Amounts (59%)
PhonePe · Payments	○	○	○	○	○	○	0 Low	Failed Transfers & Debited Amounts (54%)

High: Swiggy (4), Flipkart, Amazon, Paytm (3). **Medium:** Blinkit, Meesho, Domino's, Myntra. **Low:** Zomato, Google Pay, PhonePe.

Recommendations, Each Backed by a Number

Food & Grocery

Fix first: Fix the refund and cancellation flow: 35% of complaints (28,656), in every food app.

Most damaging: Food quality is the angriest topic (−0.44, 59% strongly negative): act fast on stale or wrong food.

Shopping

Fix first: Fix return rejections and defective products: 25% of complaints, largest in Flipkart, Meesho and Myntra.

Most damaging: Answer scam allegations quickly and visibly: the most hostile reviews (−0.61).

Payments

Fix first: Resolve debited-but-not-received payments: 52% of complaints.

Most damaging: Support that does not answer drives the anger (−0.23), more than the failure itself (−0.15).

Swiggy · High (4)

- Fix refunds first (37% of its complaints)
- Review 4.107.1: support complaints +8.9 pts
- Plan for ~1,037 1–2★ reviews next week

Flipkart · High (3)

- Fix return rejections (25%)
- Daily alert: 18 days above the forecast, the most
- Review 9.3: support and returns rose

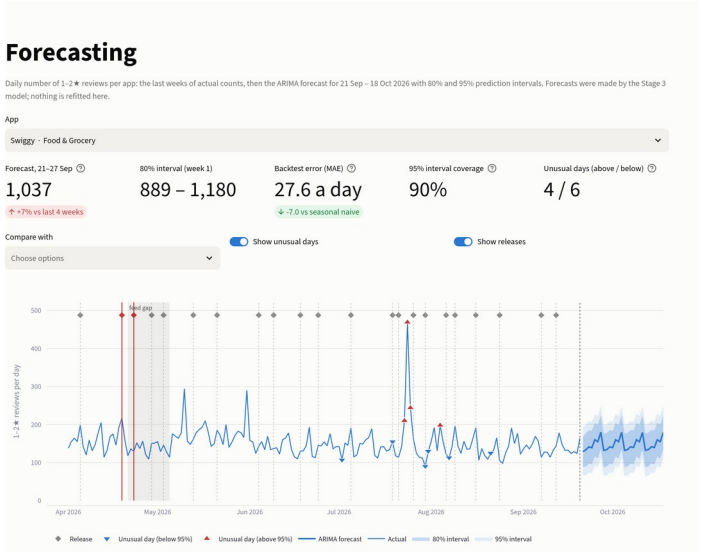
Amazon · High (3)

- Fix escalations in customer service (24%)
- 53% of its reviews are 1–2★, the highest share
- Classifier triage works best here (91% precision)

Paytm · High (3)

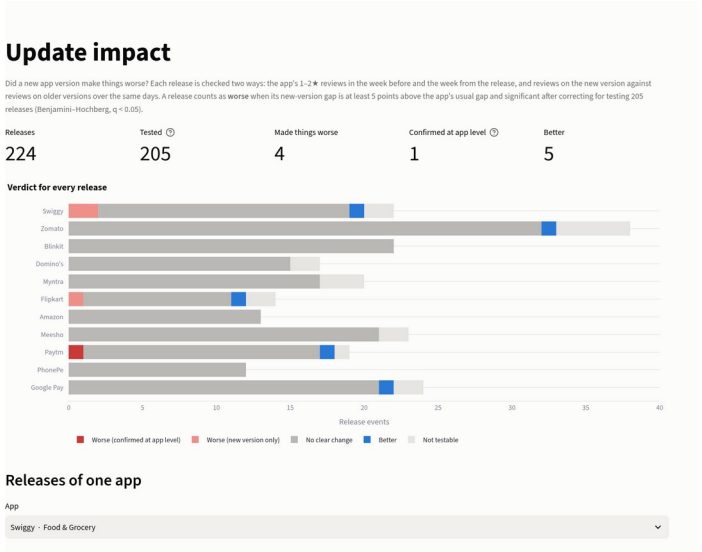
- Investigate 10.84.5 first: 34 → 66 a day
- Plan on last week's level, not the forecast
- Human check on flags (precision 66%)

Dashboard: Forecasting, Update Impact and Recommendations



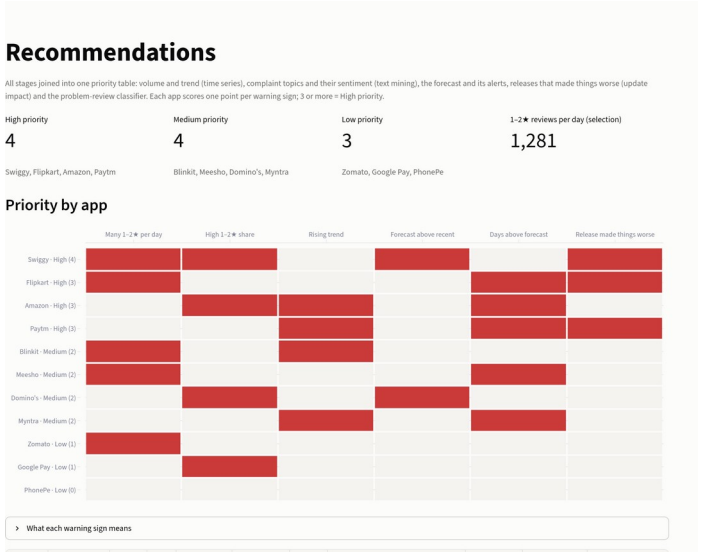
Forecasting

ARIMA forecast with 80% and 95% intervals per app, unusual days, release markers (red = made things worse), optional baseline comparison



Update impact

verdict for every release, new vs older gap against the app's usual gap, before/after detail and issue mix of any release



Recommendations

priority counts, warning-sign grid, priority table and the 65 recommendations, filtered by domain and app, with CSV download

Project Tracking and Individual Contributions

Board: github.com/users/Adithya-Monish-Kumar-K/projects/4 **Repository:** github.com/Adithya-Monish-Kumar-K/ba-capstone-app-reviews

Backlog	To Do	In Progress	Review/Testing	Completed: 14 of 14 tasks

Member	Stage	Board tasks (all due 11 Oct, all Completed)	Pull requests (merged)	Evidence
Aditya Monish Kumar K - CB.SC.U4CSE23103	1 and 6	#131 time series and patterns, #132 text preprocessing, #133 dashboard skeleton and overview; Stage 6: deck, final report, contribution summary	#147, #148, #149 (8 Oct)	scripts 15–16, dashboard/, FINAL_REPORT.md
Regella Krishna Saketh - CB.SC.U4CSE23649	2	#134 topic modelling and distinctive terms, #136 topic evaluation and interpretation	#150, #151 (8 Oct)	scripts 17–18, models/textmining/, TEXT_MINING.md §2–3
Akshay KS - CB.SC.U4CSE23104	3	#135 ARIMA forecasting with rolling backtest, #139 text-mining dashboard page	#152, #153 (9 Oct)	scripts/19_forecasting.py, 20_topic_examples.py, TIME_SERIES.md §2, text_mining.py
Harshini Vennela - CB.SC.U4CSE23455	4	#137 forecast evaluation and alerts, #142 forecast write-up, #146 check-a-review page	#154, #155, #156 (10 Oct)	scripts/20_forecast_evaluation.py, TIME_SERIES.md §2.5, check_review.py
Kanishka D - CB.SC.U4CSE23155	5	#138 update impact, #141 combined insights, #140 three dashboard pages, #143 final notebook	#157, #158 (11 Oct)	scripts 21–22, COMBINED_INSIGHTS.md, notebooks/Review_2.ipynb

Activity history: one stage after another, each merged before the next began (Stages 1–2 on 8 Oct, Stage 3 on 9 Oct, Stage 4 on 10 Oct, Stage 5 on 11 Oct). Each task is linked to a merged pull request, and every card is in Completed.

Conclusions, Limitations and Submission Links

What we found

- Complaints are mostly about service, not the app: refunds, returns, failed transfers and support lead every domain.
- Topic models cover the 43% of complaints that keyword rules missed.
- ARIMA forecasts 1–2★ reviews better than the baseline for 10 of 11 apps, with honest intervals.
- Most releases change nothing; Paytm 10.84.5 clearly made things worse.
- Fix first: Swiggy, Flipkart, Amazon and Paytm.

Limitations

- 173 days of data: no yearly pattern, only 9 backtest origins.
- A 1–2★ review is a user's judgement, not a confirmed defect.
- VADER underrates calm technical complaints; topic names are our reading of the top terms.
- Users choose when to update, and three of the worse releases overlap the feed gap.
- Counts follow app traffic and campaigns that the models do not see.

Submission links

Live dashboard
team10-app-reviews.streamlit.app

Final notebook
notebooks/Review_2.ipynb

Final report
FINAL_REPORT.md

Contribution summary
docs/Review_2_Contribution_Summary.md

Project board
github.com/users/.../projects/4

Repository
ba-capstone-app-reviews